

GVPT Data Science Camp

Data Manipulation

Ryan Bookstein

August 26, 2026

University of Maryland

- ▶ Understanding basic operations in R

Learning Objectives

- ▶ Understanding basic operations in R
- ▶ Introduction to `dplyr`

Learning Objectives

- ▶ Understanding basic operations in R
- ▶ Introduction to `dplyr`
- ▶ Cleaning and manipulating data

Objects and Functions

- ▶ Create new objects with `<-`

```
x <- 6 * 8
```

```
x
```

```
x <- 2 * 15
```

```
x
```

- ▶ Several functions come pre-installed:

```
seq(15, 30)
```

- ▶ Several functions come pre-installed:

```
seq(15, 30)
```

- ▶ You can use functions to create objects:

```
x <- seq(15, 30)
```

```
x
```


Introduction to Data Manipulation

What is Data Manipulation?

- ▶ Data rarely arrive in exactly the form you need for analysis

What is Data Manipulation?

- ▶ Data rarely arrive in exactly the form you need for analysis
- ▶ Data manipulation is reshaping data to answer your question: picking out rows and columns, creating new variables, and computing summaries

What is Data Manipulation?

- ▶ Data rarely arrive in exactly the form you need for analysis
- ▶ Data manipulation is reshaping data to answer your question: picking out rows and columns, creating new variables, and computing summaries
- ▶ Today's tool: the `dplyr` package from the tidyverse

- ▶ Load the packages we used yesterday:

```
library(tidyverse)  
library(readxl)
```

- ▶ Load the packages we used yesterday:

```
library(tidyverse)
library(readxl)
```

- ▶ Re-run yesterday's setup code (`read_excel()`, `rename()`, `select()`, and `mutate()`) to re-create the `cel` object, then view the data:

```
view(cel)
```

► In `cel`:

- ▶ In `cel`:
- ▶ `fct` stands for factors, which `R` uses to represent categorical variables with fixed values — like `benchmark` (Below, Meets, Exceeds)

- ▶ In `cel`:
- ▶ `fct` stands for factors, which `R` uses to represent categorical variables with fixed values — like `benchmark` (Below, Meets, Exceeds)
- ▶ `dbl` stands for doubles — like `les` and `ideology`

- ▶ In `cel`:
- ▶ `fct` stands for factors, which `R` uses to represent categorical variables with fixed values — like `benchmark` (Below, Meets, Exceeds)
- ▶ `dbl` stands for doubles — like `les` and `ideology`
- ▶ `chr` stands for strings (character vectors) — like `member`, `state`, and `party`

► `int` stands for integer

- ▶ `int` stands for integer
- ▶ `dtm` stands for date-times (a date + time)

- ▶ `int` stands for integer
- ▶ `dtm` stands for date-times (a date + time)
- ▶ `lgl` stands for logical, vectors that contain only `TRUE` or `FALSE`

- ▶ Five basic functions:

- ▶ Five basic functions:

- ▷ `filter()`

► Five basic functions:

- ▷ `filter()`
- ▷ `arrange()`

► Five basic functions:

- ▷ `filter()`
- ▷ `arrange()`
- ▷ `select()`

► Five basic functions:

- ▷ `filter()`
- ▷ `arrange()`
- ▷ `select()`
- ▷ `mutate()`

► Five basic functions:

- ▷ `filter()`
- ▷ `arrange()`
- ▷ `select()`
- ▷ `mutate()`
- ▷ `summarise()`

Filtering and Arranging Data

Filtering Rows

```
filter(cel, state == "MD", congress > 110)
```

```
# A tibble: 64 x 14
```

	member	congress	year	state	dem	female	votepct	ideology
	<chr>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	Bartlett, Roscoe	111	2009	MD	0	0	58	0.493
2	Cummings, Elijah	111	2009	MD	1	0	80	-0.438
3	Edwards, Donna	111	2009	MD	1	1	86	-0.562
4	Hoyer, Steny	111	2009	MD	1	0	74	-0.380
5	Kratovil, Frank	111	2009	MD	1	0	49	-0.0370
6	Ruppersberger, C.	111	2009	MD	1	0	72	-0.295

```
# i 58 more rows
```

```
# i 6 more variables: seniority <dbl>, born <dbl>, les <dbl>,
```

```
# bills_passed <dbl>, benchmark <fct>, party <chr>
```

- ▶ First argument must always be a data frame

- ▶ First argument must always be a data frame
- ▶ The arguments that follow describe which columns to operate on, using the variable names (without quotes)

- ▶ First argument must always be a data frame
- ▶ The arguments that follow describe which columns to operate on, using the variable names (without quotes)
- ▶ Output is always a new data frame

Filtering Rows

```
filter(cel, state %in% c("MD", "VA"))
```

```
# A tibble: 492 x 14
```

	member <chr>	congress <dbl>	year <dbl>	state <chr>	dem <dbl>	female <dbl>	votepct <dbl>	ideology <dbl>
1	Bauman, Robert	93	1973	MD	0	0	NA	0.533
2	Broyhill, Joel	93	1973	VA	0	0	56	0.159
3	Butler, M.	93	1973	VA	0	0	55	0.332
4	Byron, Goodloe	93	1973	MD	1	0	65	-0.0140
5	Daniel, Robert	93	1973	VA	0	0	47	0.318
6	Daniel, W.	93	1973	VA	1	0	100	0.161

```
# i 486 more rows
```

```
# i 6 more variables: seniority <dbl>, born <dbl>, les <dbl>,
```

```
# bills_passed <dbl>, benchmark <fct>, party <chr>
```

Filtering Rows

```
filter(cel, vote_pct > 50 & vote_pct < 60)
```

```
# A tibble: 3,023 x 14
```

	member	congress	year	state	dem	female	vote_pct	ideology
	<chr>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	Abdnor, James	93	1973	SD	0	0	55	0.241
2	Abzug, Bella	93	1973	NY	1	1	56	-0.597
3	Annunzio, Frank	93	1973	IL	1	0	53	-0.398
4	Arends, Leslie	93	1973	IL	0	0	57	0.301
5	Ashbrook, John	93	1973	OH	0	0	57	0.612
6	Baker, LaMar	93	1973	TN	0	0	55	0.308

```
# i 3,017 more rows
```

```
# i 6 more variables: seniority <dbl>, born <dbl>, les <dbl>,
```

```
# bills_passed <dbl>, benchmark <fct>, party <chr>
```

Filtering Rows

```
filter(cel, vote_pct < 50 | vote_pct > 60)
```

```
# A tibble: 7,617 x 14
```

	member <chr>	congress <dbl>	year <dbl>	state <chr>	dem <dbl>	female <dbl>	vote_pct <dbl>	ideology <dbl>
1	Adams, Brock	93	1973	WA	1	0	85	-0.396
2	Addabbo, Joseph	93	1973	NY	1	0	75	-0.402
3	Albert, Carl	93	1973	OK	1	0	93	-0.392
4	Alexander, Bill	93	1973	AR	1	0	100	-0.410
5	Anderson, John	93	1973	IL	0	0	72	0.183
6	Anderson, Glenn	93	1973	CA	1	0	75	-0.269

```
# i 7,611 more rows
```

```
# i 6 more variables: seniority <dbl>, born <dbl>, les <dbl>,
```

```
#   bills_passed <dbl>, benchmark <fct>, party <chr>
```

- ▶ Find all member-Congress observations where the member ran unopposed (received 100% of the vote)

- ▶ Find all member-Congress observations where the member ran unopposed (received 100% of the vote)
- ▶ Find all members from Maryland

- ▶ Find all member-Congress observations where the member ran unopposed (received 100% of the vote)
- ▶ Find all members from Maryland
- ▶ Find all members from Maryland and Virginia

- ▶ Find all member-Congress observations where the member ran unopposed (received 100% of the vote)
- ▶ Find all members from Maryland
- ▶ Find all members from Maryland and Virginia
- ▶ Find all member-Congress observations with a vote share greater than 60% and less than 70%

- ▶ Find all member-Congress observations where the member ran unopposed (received 100% of the vote)
- ▶ Find all members from Maryland
- ▶ Find all members from Maryland and Virginia
- ▶ Find all member-Congress observations with a vote share greater than 60% and less than 70%
- ▶ Find all member-Congress observations with a vote share less than 60% or greater than 70%

Check Your Answers

```
filter(cel, vote_pct == 100)
```

```
# A tibble: 750 x 14
```

	member <chr>	congress <dbl>	year <dbl>	state <chr>	dem <dbl>	female <dbl>	vote_pct <dbl>	ideology <dbl>
1	Alexander, Bill	93	1973	AR	1	0	100	-0.410
2	Boland, Edward	93	1973	MA	1	0	100	-0.306
3	Breaux, John	93	1973	LA	1	0	100	-0.123
4	Brinkley, Jack	93	1973	GA	1	0	100	-0.0730
5	Burke, James	93	1973	MA	1	0	100	-0.400
6	Burleson, Omar	93	1973	TX	1	0	100	0.00200

```
# i 744 more rows  
# i 6 more variables: seniority <dbl>, born <dbl>, les <dbl>,  
#   bills_passed <dbl>, benchmark <fct>, party <chr>
```

Check Your Answers

```
filter(cel, state == "MD")
```

```
# A tibble: 211 x 14
```

	member <chr>	congress <dbl>	year <dbl>	state <chr>	dem <dbl>	female <dbl>	votepct <dbl>	ideology <dbl>
1	Bauman, Robert	93	1973	MD	0	0	NA	0.533
2	Byron, Goodloe	93	1973	MD	1	0	65	-0.0140
3	Gude, Gilbert	93	1973	MD	0	0	64	-0.0280
4	Hogan, Lawrence	93	1973	MD	0	0	63	0.158
5	Holt, Marjorie	93	1973	MD	0	1	59	0.368
6	Long, Clarence	93	1973	MD	1	0	66	-0.274

```
# i 205 more rows  
# i 6 more variables: seniority <dbl>, born <dbl>, les <dbl>,  
#   bills_passed <dbl>, benchmark <fct>, party <chr>
```

Check Your Answers

```
filter(cel, state %in% c("MD", "VA"))
```

```
# A tibble: 492 x 14
```

	member <chr>	congress <dbl>	year <dbl>	state <chr>	dem <dbl>	female <dbl>	voteptct <dbl>	ideology <dbl>
1	Bauman, Robert	93	1973	MD	0	0	NA	0.533
2	Broyhill, Joel	93	1973	VA	0	0	56	0.159
3	Butler, M.	93	1973	VA	0	0	55	0.332
4	Byron, Goodloe	93	1973	MD	1	0	65	-0.0140
5	Daniel, Robert	93	1973	VA	0	0	47	0.318
6	Daniel, W.	93	1973	VA	1	0	100	0.161

```
# i 486 more rows
```

```
# i 6 more variables: seniority <dbl>, born <dbl>, les <dbl>,
```

```
# bills_passed <dbl>, benchmark <fct>, party <chr>
```

Check Your Answers

```
filter(cel, vote_pct > 60 & vote_pct < 70)
```

```
# A tibble: 3,423 x 14
```

	member <chr>	congress <dbl>	year <dbl>	state <chr>	dem <dbl>	female <dbl>	vote_pct <dbl>	ideology <dbl>
1	Armstrong, William	93	1973	CO	0	0	62	0.508
2	Ashley, Thomas	93	1973	OH	1	0	69	-0.350
3	Aspin, Les	93	1973	WI	1	0	64	-0.320
4	Bafalis, L.	93	1973	FL	0	0	62	0.303
5	Barrett, William	93	1973	PA	1	0	66	-0.469
6	Bell, Alphonzo	93	1973	CA	0	0	61	0.183

```
# i 3,417 more rows  
# i 6 more variables: seniority <dbl>, born <dbl>, les <dbl>,  
#   bills_passed <dbl>, benchmark <fct>, party <chr>
```

Check Your Answers

```
filter(cel, vote_pct < 60 | vote_pct > 70)
```

```
# A tibble: 7,142 x 14
```

	member <chr>	congress <dbl>	year <dbl>	state <chr>	dem <dbl>	female <dbl>	vote_pct <dbl>	ideology <dbl>
1	Abdnor, James	93	1973	SD	0	0	55	0.241
2	Abzug, Bella	93	1973	NY	1	1	56	-0.597
3	Adams, Brock	93	1973	WA	1	0	85	-0.396
4	Addabbo, Joseph	93	1973	NY	1	0	75	-0.402
5	Albert, Carl	93	1973	OK	1	0	93	-0.392
6	Alexander, Bill	93	1973	AR	1	0	100	-0.410

```
# i 7,136 more rows  
# i 6 more variables: seniority <dbl>, born <dbl>, les <dbl>,  
#   bills_passed <dbl>, benchmark <fct>, party <chr>
```

▶ `==` is equal to

Important Features

- ▶ `==` is equal to
- ▶ `!=` is not equal to

Important Features

- ▶ `==` is equal to
- ▶ `!=` is not equal to
- ▶ `>` is greater than, `<` is less than

Important Features

- ▶ `==` is equal to
- ▶ `!=` is not equal to
- ▶ `>` is greater than, `<` is less than
- ▶ `>=` is greater than or equal to

Important Features

- ▶ `==` is equal to
- ▶ `!=` is not equal to
- ▶ `>` is greater than, `<` is less than
- ▶ `>=` is greater than or equal to
- ▶ `<=` is less than or equal to

▶ | is or

Important Features

- ▶ `|` is or
- ▶ `&` is and

Important Features

- ▶ `|` is or
- ▶ `&` is and
- ▶ `%in%` is in

Arrange Rows with arrange()

```
arrange(cel, member, congress)
```

```
# A tibble: 11,606 x 14
```

	member <chr>	congress <dbl>	year <dbl>	state <chr>	dem <dbl>	female <dbl>	votepct <dbl>	ideology <dbl>
1	Abdnor, James	93	1973	SD	0	0	55	0.241
2	Abdnor, James	94	1975	SD	0	0	68	0.241
3	Abdnor, James	95	1977	SD	0	0	70	0.241
4	Abdnor, James	96	1979	SD	0	0	NA	0.241
5	Abercrombie, Neil	99	1985	HI	1	0	NA	-0.432
6	Abercrombie, Neil	102	1991	HI	1	0	60	-0.432

```
# i 11,600 more rows  
# i 6 more variables: seniority <dbl>, born <dbl>, les <dbl>,  
#   bills_passed <dbl>, benchmark <fct>, party <chr>
```

Arrange Rows with arrange()

```
arrange(cel, member, desc(congress))
```

```
# A tibble: 11,606 x 14
```

	member	congress	year	state	dem	female	votepct	ideology
	<chr>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	Abdnor, James	96	1979	SD	0	0	NA	0.241
2	Abdnor, James	95	1977	SD	0	0	70	0.241
3	Abdnor, James	94	1975	SD	0	0	68	0.241
4	Abdnor, James	93	1973	SD	0	0	55	0.241
5	Abercrombie, Neil	111	2009	HI	1	0	77	-0.432
6	Abercrombie, Neil	110	2007	HI	1	0	69	-0.432

```
# i 11,600 more rows  
# i 6 more variables: seniority <dbl>, born <dbl>, les <dbl>,  
#   bills_passed <dbl>, benchmark <fct>, party <chr>
```

- ▶ Which member-Congress observations have the highest Legislative Effectiveness Score (**les**)?

- ▶ Which member-Congress observations have the highest Legislative Effectiveness Score (`les`)?
- ▶ Which member-Congress observation passed the most bills through the House (`bills_passed`)?

Check Your Answers

```
arrange(cel, desc(les)) |>
  select(member, congress, les, bills_passed)
```

```
# A tibble: 11,606 x 4
  member          congress    les bills_passed
  <chr>          <dbl> <dbl>         <dbl>
1 Rangel, Charles    110  18.7           33
2 Murphy, John       96  17.9           31
3 Sullivan, Leonor   93  17.6           28
4 Sensenbrenner, F.  109  17.6           25
5 Murphy, John       95  17.2           31
6 Ullman, Al         96  16.4           18
7 Smith, Lamar      112  16.3           20
8 Young, Don        108  16.3           20
# i 11,598 more rows
```

Check Your Answers

```
arrange(cel, desc(bills_passed)) |>
  select(member, congress, les, bills_passed)
```

```
# A tibble: 11,606 x 4
  member          congress    les bills_passed
  <chr>          <dbl> <dbl>         <dbl>
1 Rangel, Charles    110  18.7           33
2 Murphy, John       95  17.2           31
3 Murphy, John       96  17.9           31
4 Sullivan, Leonor   93  17.6           28
5 Sullivan, Leonor   94  12.7           25
6 Sensenbrenner, F.  109  17.6           25
7 Young, Don         107  15.8           22
8 Miller, George     110  10.6           22
# i 11,598 more rows
```

Choosing and Creating Columns

Select Columns with select()

```
select(cel, member, congress, les)
```

```
# A tibble: 11,606 x 3
  member      congress    les
  <chr>      <dbl>    <dbl>
1 Abdnor, James      93 0.110
2 Abzug, Bella       93 0.762
3 Adams, Brock       93 1.24
4 Addabbo, Joseph    93 0.155
5 Albert, Carl       93 0.00103
6 Alexander, Bill    93 1.88
7 Anderson, John     93 0.289
8 Anderson, Glenn    93 0.447
# i 11,598 more rows
```

Select Columns with select()

```
select(cel, member:state)
```

```
# A tibble: 11,606 x 4
  member      congress  year state
  <chr>          <dbl> <dbl> <chr>
1 Abdnor, James      93  1973 SD
2 Abzug, Bella       93  1973 NY
3 Adams, Brock       93  1973 WA
4 Addabbo, Joseph    93  1973 NY
5 Albert, Carl       93  1973 OK
6 Alexander, Bill    93  1973 AR
7 Anderson, John     93  1973 IL
8 Anderson, Glenn    93  1973 CA
# i 11,598 more rows
```

Select Columns with select()

```
select(cel, -(votepct:ideology))
```

```
# A tibble: 11,606 x 12
  member      congress year state  dem female seniority  born
  les
  <chr>          <dbl> <dbl> <chr> <dbl>  <dbl>      <dbl> <dbl>
  <dbl>
1 Abdnor, James      93  1973 SD      0      0          1  1923
  0.110
2 Abzug, Bella       93  1973 NY      1      1          2  1920
  0.762
3 Adams, Brock       93  1973 WA      1      0          5  1927 1.24
4 Addabbo, Joseph   93  1973 NY      1      0          7  1925
  0.155
# i 11,602 more rows
# i 3 more variables: bills_passed <dbl>, benchmark <fct>, party <chr>
```

- ▶ Select only the `member`, `congress`, and `votepct` variables from `cel`

Exercise

- ▶ Select only the `member`, `congress`, and `votepct` variables from `cel`
- ▶ What does the `any_of()` function do? Why might it be useful with this vector?

```
vars <- c("member", "congress", "votepct")
```

- ▶ Select only the `member`, `congress`, and `votepct` variables from `cel`
- ▶ What does the `any_of()` function do? Why might it be useful with this vector?

```
vars <- c("member", "congress", "votepct")
```

- ▶ What does this code produce?

```
select(cel, starts_with("b"))
```

Check Your Answers

```
select(cel, member, congress, vote_pct)
```

```
# A tibble: 11,606 x 3
  member      congress vote_pct
  <chr>      <dbl>    <dbl>
1 Abdnor, James      93      55
2 Abzug, Bella       93      56
3 Adams, Brock       93      85
4 Addabbo, Joseph    93      75
5 Albert, Carl       93      93
6 Alexander, Bill    93     100
7 Anderson, John     93      72
8 Anderson, Glenn    93      75
# i 11,598 more rows
```

Check Your Answers

`any_of()` selects any of the named columns that exist, without erroring on ones that don't:

```
vars <- c("member", "congress", "votepct")  
  
select(cel, any_of(vars))
```

```
# A tibble: 11,606 x 3  
  member      congress votepct  
  <chr>      <dbl>    <dbl>  
1 Abdnor, James      93      55  
2 Abzug, Bella       93      56  
3 Adams, Brock       93      85  
4 Addabbo, Joseph   93      75  
5 Albert, Carl      93      93  
6 Alexander, Bill   93     100  
# i 11,600 more rows
```

Check Your Answers

All columns whose names start with “b”: **born**, **bills_passed**, and **benchmark**

```
select(cel, starts_with("b"))
```

```
# A tibble: 11,606 x 3
  born bills_passed benchmark
<dbl>         <dbl> <fct>
1  1923             0 Below
2  1920             1 Meets
3  1927             2 Meets
4  1925             0 Below
5  1908             0 Below
6  1934             2 Meets
7  1922             0 Meets
8  1913             0 Meets
# i 11,598 more rows
```

Add New Variables with mutate()

```
mutate(cel, age = year - born) |>
  select(member, congress, year, born, age)
```

```
# A tibble: 11,606 x 5
  member      congress  year  born  age
  <chr>      <dbl> <dbl> <dbl> <dbl>
1 Abdnor, James      93  1973  1923   50
2 Abzug, Bella       93  1973  1920   53
3 Adams, Brock       93  1973  1927   46
4 Addabbo, Joseph    93  1973  1925   48
5 Albert, Carl       93  1973  1908   65
6 Alexander, Bill    93  1973  1934   39
7 Anderson, John     93  1973  1922   51
8 Anderson, Glenn    93  1973  1913   60
# i 11,598 more rows
```

Select, Manipulate, and Add New Variables with transmute()

```
transmute(cel, member, congress, age = year - born)
```

```
# A tibble: 11,606 x 3
  member      congress  age
  <chr>      <dbl> <dbl>
1 Abdnor, James      93    50
2 Abzug, Bella       93    53
3 Adams, Brock       93    46
4 Addabbo, Joseph    93    48
5 Albert, Carl       93    65
6 Alexander, Bill    93    39
7 Anderson, John     93    51
8 Anderson, Glenn    93    60
# i 11,598 more rows
```

Summaries and the Pipe

Create Summaries with summarise()

```
summarise(cel, avg_bills = mean(bills_passed),  
          avg_seniority = mean(seniority))
```

```
# A tibble: 1 x 2  
  avg_bills avg_seniority  
    <dbl>      <dbl>  
1     1.48         5.27
```

Creating Grouped Summaries with `group_by()` and `summarise()`

```
cel_party <- group_by(cel, party)

summarise(cel_party, avg_bills = mean(bills_passed),
          avg_seniority = mean(seniority))
```

```
# A tibble: 2 x 3
  party      avg_bills avg_seniority
  <chr>      <dbl>      <dbl>
1 Democrat    1.57        5.75
2 Republican  1.38        4.72
```

- ▶ Calculate each member's age in each Congress

- ▶ Calculate each member's age in each Congress
- ▶ Find each member's average number of bills passed across all Congresses in the `cel` dataset

- ▶ Calculate each member's age in each Congress
- ▶ Find each member's average number of bills passed across all Congresses in the `cel` dataset
- ▶ Find the member with the lowest average bills passed across these Congresses (careful: many members tie at zero)

- ▶ Calculate each member's age in each Congress
- ▶ Find each member's average number of bills passed across all Congresses in the `cel` dataset
- ▶ Find the member with the lowest average bills passed across these Congresses (careful: many members tie at zero)
- ▶ Find the member with the highest average bills passed across these Congresses

Check Your Answers

```
mutate(cel, age = year - born) |>
  select(member, congress, year, born, age)
```

```
# A tibble: 11,606 x 5
  member      congress year  born  age
  <chr>      <dbl> <dbl> <dbl> <dbl>
1 Abdnor, James      93  1973  1923   50
2 Abzug, Bella       93  1973  1920   53
3 Adams, Brock       93  1973  1927   46
4 Addabbo, Joseph    93  1973  1925   48
5 Albert, Carl       93  1973  1908   65
6 Alexander, Bill    93  1973  1934   39
7 Anderson, John     93  1973  1922   51
8 Anderson, Glenn    93  1973  1913   60
# i 11,598 more rows
```

Check Your Answers

```
cel_member <- group_by(cel, member)

avg_bills_by_member <- summarise(cel_member,
  avg_bills = mean(bills_passed))

avg_bills_by_member
```

```
# A tibble: 2,233 x 2
  member          avg_bills
  <chr>          <dbl>
1 Abdnor, James    0.5
2 Abercrombie, Neil 0.545
3 Abraham, Ralph   1.33
4 Abzug, Bella     1.5
5 Acevedo-Vila, Anibal 2
# i 2,228 more rows
```


Check Your Answers

```
arrange(avg_bills_by_member, avg_bills)
```

```
# A tibble: 2,233 x 2
  member          avg_bills
  <chr>          <dbl>
1 Albert, Carl      0
2 Ammerman, Joseph  0
3 Amo, Gabe         0
4 Arends, Leslie    0
5 Ashbrook, John    0
6 Austria, Steve    0
7 Bacchus, Jim      0
8 Badillo, Herman   0
# i 2,225 more rows
```

Check Your Answers

```
arrange(avg_bills_by_member, desc(avg_bills))
```

```
# A tibble: 2,233 x 2
  member          avg_bills
  <chr>          <dbl>
1 Sullivan, Leonor    26.5
2 Rogers, Paul       19.7
3 Murphy, John       17
4 Ullman, Al         14.8
5 Rodino, Peter      12.1
6 Mahon, George      12
7 Hays, Wayne        11
8 Katko, John       10.5
# i 2,225 more rows
```

- ▶ We created a lot of different objects during our calculations

Utilize Multiple Operations with the Pipe

- ▶ We created a lot of different objects during our calculations
- ▶ Can we avoid creating unneeded objects?

Utilize Multiple Operations with the Pipe

- ▶ We created a lot of different objects during our calculations
- ▶ Can we avoid creating unneeded objects?
- ▶ We can utilize a feature of the tidyverse: the pipe

- ▶ Pipe can be read as:

```
cel |>  
  group_by(state) |>  
  summarise(avg_les = mean(les)) |>  
  arrange(desc(avg_les))
```

Utilize Multiple Operations with the Pipe

- ▶ Pipe can be read as:
- ▶ Take this `|>` (and then...)

```
cel |>  
  group_by(state) |>  
  summarise(avg_les = mean(les)) |>  
  arrange(desc(avg_les))
```

Utilize Multiple Operations with the Pipe

- ▶ Pipe can be read as:
- ▶ Take this `|>` (and then...)
- ▶ do this `|>` (and then...)

```
cel |>  
  group_by(state) |>  
  summarise(avg_les = mean(les)) |>  
  arrange(desc(avg_les))
```


Utilize Multiple Operations with the Pipe

- ▶ Pipe can be read as:
- ▶ Take this `|>` (and then...)
- ▶ do this `|>` (and then...)
- ▶ do this

```
cel |>  
  group_by(state) |>  
  summarise(avg_les = mean(les)) |>  
  arrange(desc(avg_les))
```

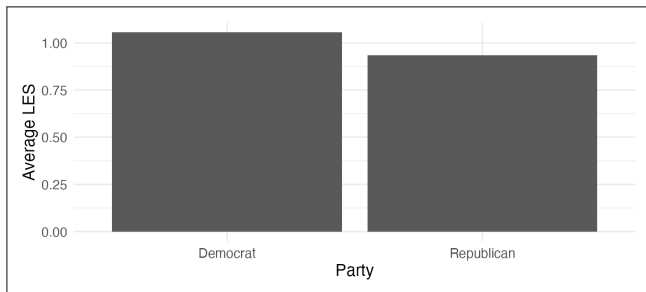
Utilize Multiple Operations with the Pipe

```
cel |>
  group_by(state) |>
  summarise(avg_les = mean(les)) |>
  arrange(desc(avg_les))
```

```
# A tibble: 56 x 2
  state avg_les
  <chr>   <dbl>
1 AK      4.87
2 DC      2.65
3 OR      1.54
4 MT      1.52
5 UT      1.50
6 NM      1.49
7 NJ      1.40
# i 49 more rows
```

Utilize Multiple Operations with the Pipe

```
cel |>  
  group_by(parties) |>  
  summarise(avg_les = mean(les)) |>  
  ggplot(aes(x = parties, y = avg_les)) +  
  geom_col() +  
  theme_minimal()
```



A Note on the Pipe

► Base pipe: `|>`

A Note on the Pipe

- ▶ Base pipe: `|>`
 - ▷ Can be used without loading any packages

- ▶ Base pipe: `|>`
 - ▷ Can be used without loading any packages
 - ▷ Relatively new: introduced in 2021

A Note on the Pipe

- ▶ Tidyverse pipe: `%>%`

A Note on the Pipe

- ▶ Tidyverse pipe: `%>%`
 - ▷ Must load `dplyr` or `magrittr` to use

- ▶ Calculate the average Legislative Effectiveness Score for Maryland members serving in Congresses after the 110th (2009 onward). Display only the member and average LES variables

- ▶ Calculate the average Legislative Effectiveness Score for Maryland members serving in Congresses after the 110th (2009 onward). Display only the member and average LES variables
- ▶ Plot the results

Check Your Answers

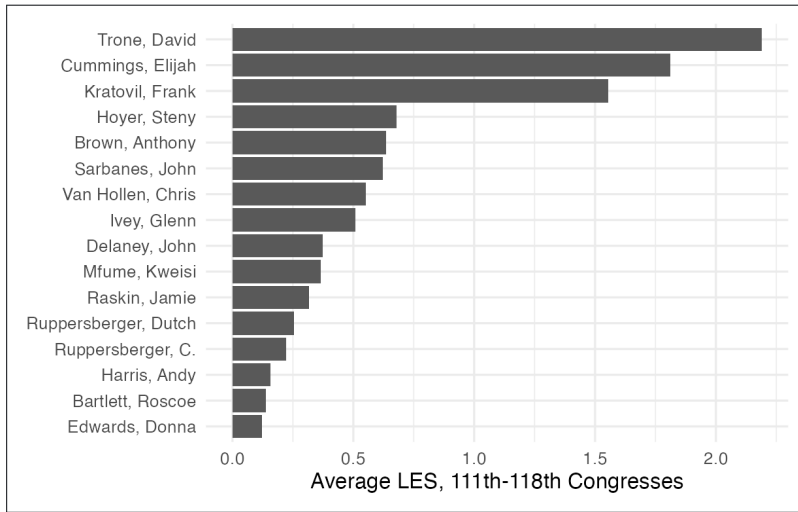
```
cel |>
  filter(state == "MD", congress > 110) |>
  group_by(member) |>
  summarise(avg_les = mean(les))
```

```
# A tibble: 16 x 2
  member          avg_les
  <chr>          <dbl>
1 Bartlett, Roscoe 0.139
2 Brown, Anthony  0.634
3 Cummings, Elijah 1.81
4 Delaney, John    0.374
5 Edwards, Donna   0.122
6 Harris, Andy     0.156
7 Hoyer, Steny     0.679
# i 9 more rows
```

- ▶ `fct_reorder()` sorts the bars: it reorders the `member` categories by `avg_les`, so the chart reads top to bottom

```
cel |>
  filter(state == "MD", congress > 110) |>
  group_by(member) |>
  summarise(avg_les = mean(les)) |>
  ggplot(aes(x = avg_les, y = fct_reorder(member, avg_les))) +
  geom_col() +
  theme_minimal() +
  labs(x = "Average LES, 111th-118th Congresses", y = NULL)
```

Check Your Answers



- ▶ Familiarized yourself with basic R syntax

- ▶ Familiarized yourself with basic R syntax
- ▶ Learned how to manipulate your data

- ▶ Familiarized yourself with basic R syntax
- ▶ Learned how to manipulate your data
- ▶ Wrote concise code that is easy to follow

- ▶ On Thursday, you will apply the skills learned in this course to an idea that interests you. In doing so, you will need to find a dataset that:

- ▶ On Thursday, you will apply the skills learned in this course to an idea that interests you. In doing so, you will need to find a dataset that:
- ▶ Is relevant to your research interests

- ▶ On Thursday, you will apply the skills learned in this course to an idea that interests you. In doing so, you will need to find a dataset that:
- ▶ Is relevant to your research interests
- ▶ Contains continuous and categorical variables

- ▶ On Thursday, you will apply the skills learned in this course to an idea that interests you. In doing so, you will need to find a dataset that:
- ▶ Is relevant to your research interests
- ▶ Contains continuous and categorical variables
- ▶ If needed, reach out to me for dataset recommendations

Questions?